Re-Implementing CNN Price-Trend Signals: A Reproduction of Jiang, Kelly, and Xiu (2023)

Kaibiao Zhu
The Hong Kong University of Science and Technology (Guangzhou)
kzhu597@connect.hkust-gz.edu.cn

Abstract

This paper reproduces the image-based price-trend prediction framework of Jiang et al. [2023] (hereafter the *original paper*), which replaces predefined technical indicators with OHLC-volume charts and convolutional neural networks (CNNs). Using CRSP daily U.S. equity data, we reconstruct the image pipeline, train horizon-specific CNN ensembles, and evaluate the resulting signals through out-of-sample backtests. We compare the CNN signals with momentum, moving-average, and reversal benchmarks, and assess residual performance through factor-spanning regressions. The reproduction is strongest at short horizons, where image-based signals deliver the best economic performance. Relative to the magnitudes reported by the original paper, medium- and long-horizon strategies weaken substantially after costs, and factor stripping yields significantly negative alphas. Saliency maps indicate that the CNN mainly attends to recent price movement and volume contrast. We also provide replication materials so that the empirical results can be audited and extended.

1 Introduction

Technical analysis has long suggested that the visual "shape" of a price chart conveys information beyond what is captured by scalar momentum or moving-average filters. The original paper formalises this intuition by feeding fixed-resolution open–high–low–close (OHLC) images with a volume sub-panel into a convolutional neural network and showing that the resulting forecasts dominate canonical price-trend signals out of sample. The study joins a growing empirical asset-pricing literature that uses flexible machine-learning predictors to map raw historical features into expected returns [Gu et al., 2020, Kelly et al., 2019], and it ties directly to early efforts at formalising chart patterns via kernel smoothing [Lo et al., 2000].

This paper re-implements the core pipeline from end to end and evaluates it under the same in-sample and out-of-sample calendar design as the original paper: training and validation draw from 1993–2000 with a 70/30 random sample split, and headline out-of-sample results are reported for 2001–2019. Our contributions are:

1. An independent OHLC-plus-volume image renderer and horizon-matched CNN stacks built from the dimensions, drawing rules, and layer geometry documented in the original paper—including the simple moving-average overlay, volume sub-panel, and vertical orientation convention—implemented from the original paper's written specification and reported tensor dimensions.
2. A CRSP-style daily panel for data construction, with headline out-of-sample CNN evaluation as above; proportional transaction costs; matched-horizon price momentum, simple moving-average cross, and short-term reversal baselines; and a five-factor-plus-momentum spanning regression [Fama and French, 1993, 2015, Carhart, 1997].

3. Input-gradient saliency analysis [?] of the $I = 60$ ensemble, which highlights the right tail of each image and the volume bars as the dominant feature regions.
4. Code, trained weights, and post-aggregation CSV tables are released on GitHub, allowing every numerical claim to be regenerated.

The remainder of the paper is organised as follows. Section 2 positions our reproduction relative to the chart-pattern, deep-learning, and interpretability literatures. Section 3 details data sources and image construction. Section 4 describes the CNN architecture and training recipe. Section 5 reports empirical results: cross-sectional decile analyses against momentum, moving-average cross, and short-term reversal benchmarks; volatility-adjusted cumulative long–short performance (Figure 5–style); portfolio statistics; classification metrics; and Fama–French factor spanning. Section 6 visualises model attention. Section 7 discusses where our results align with and diverge from the original paper.

2 Related work

Chart-pattern literature. Lo et al. [2000] provided one of the first systematic treatments of technical chart patterns by combining kernel regression with automated pattern recognition. Jegadeesh and Titman [1993] documented price momentum as a robust cross-sectional return predictor, providing the canonical baseline against which chart-based technical trend signals are often benchmarked empirically.

Deep learning for return prediction. Recent work demonstrates that deep architectures uncover return signals that are difficult to capture with closed-form characteristics. Gu et al. [2020] benchmark a broad set of supervised learners on U.S. equities; Kelly et al. [2019] use latent-factor models with characteristic loadings. Within technical analysis, Sezer and Ozbayoglu [2018] convert indicators into images for trading. The original paper unifies these threads by treating the OHLC chart itself as the model input.

Interpretability via input gradients. **?** introduced gradient-based saliency to visualise which input pixels most influence a CNN's class score; we apply the same technique to identify which segments of each OHLC image drive the model's "up" probability. CNN architectures used here trace back to LeCun et al. [1998].

3 Data and image construction

Universe. We employ a CRSP-consistent daily security-level panel of U.S. equities spanning 1992-01-01, to 2024-12-31. Data are observed at the daily frequency for each security, uniquely identified by its CRSP permanent security number (PERMNO). The panel includes core trading and pricing fields: daily market capitalization, total and price returns, trading volume, and open-high-low-close (OHLC) prices. We impose a standard sequential screening procedure: all fields are required to be non-missing; we retain only securities with a closing price of at least $1 and strictly positive trading volume. Log returns are computed, and OHLC prices are adjusted for corporate actions via cumulative log-return compounding, with the first valid observation for each security serving as the normalization anchor. No additional survivorship bias adjustments are applied beyond those inherent to the standard CRSP universe construction.

Labels and holding period. For each horizon $F \in \{5, 20, 60\}$ trading days we form $R^{\text{fut}}_{t,F} = \exp\bigl(\sum_{s=t+1}^{t+F} \log(1 + r_s)\bigr) - 1$ on the adjusted price path, and a binary classification label $y_{t,F} = \mathbf{1}\{R^{\text{fut}}_{t,F} > 0\}$. Rebalance timestamps are fixed at period-end calendar boundaries (Friday for $F = 5$, month-end for $F = 20$, quarter-end for $F = 60$), matching the canonical SPY trade calendar.

Image construction. Following the original paper, we encode each stock–date observation as a fixed-size grayscale *chart image* rather than a vector of technical indicators. The background is black (pixel value 0) and all price–volume objects are drawn in white (255). We consider three input horizons $I \in \{5, 20, 60\}$, each matched to the prediction horizon $F = I$, so the image summarizes the most recent I trading days of adjusted OHLC prices and daily volume.

OHLC panel. The upper portion of the image is reserved for open–high–low–close prices. Each trading day τ in the I-day window occupies three adjacent pixel columns: a one-pixel open marker in the left column, a vertical high–low bar in the centre, and a one-pixel close marker in the right column. Prices are linearly mapped into the vertical axis using the window minimum and maximum, where the range is defined to include both the raw OHLC values and the moving-average line described below. After drawing, the OHLC sub-panel is flipped top-to-bottom so that upward price movements read visually as "up" on the chart, in line with conventional price plots.

Moving-average line. We overlay a simple moving average (SMA) of the close price, using an I-day window—the same length as the chart horizon (e.g., a 20-day chart uses a 20-day SMA). For each day, the SMA value is placed on the centre column of that day; consecutive days are connected by a one-pixel poly-line. To ensure that the first in-window SMA already reflects I days of history, the SMA is computed from $2I - 1$ days of past closes ending at the chart date.

Volume sub-panel. The lower portion of the image displays volume bars. As in the original paper, the volume chart is drawn along the bottom of the image and occupies one-fifth of the total height; the OHLC panel fills the upper four-fifths, with a one-pixel gap between the two regions. Within the volume band, the largest daily volume in the window is scaled to the top of the band and all other days are drawn proportionally. Missing-volume days are left blank.

Image dimensions. Because each day uses three pixel columns, the image width is $W = 3I$. The total height H stacks the OHLC block, the separator, and the volume band. Table 1 reports the (W, H) pairs used in our implementation, identical to those in the original paper.

Table 1: Image Dimensions

Horizon	W	H
$I = 5$	15	32
$I = 20$	60	64
$I = 60$	180	96

Figure 1 provides a concrete illustration of the construction rules above. We plot the same security on the same calendar anchor for $I \in \{5, 20, 60\}$, so the only deliberate difference across panels is the look-back length. The left panel ($I = 5$) is the coarsest view: fifteen columns of price action and a compact volume strip. The centre and right panels progressively widen the canvas to sixty and one hundred eighty columns, revealing more fine-grained OHLC structure, a longer SMA trace, and richer intra-window variation in volume. Because every image fed to the CNN has fixed (W, H) for a given horizon, longer windows do not add pixels per trading day; instead, the model sees the same economic history at three standardised resolutions, analogous to viewing a price chart at weekly, monthly, and quarterly zoom levels.

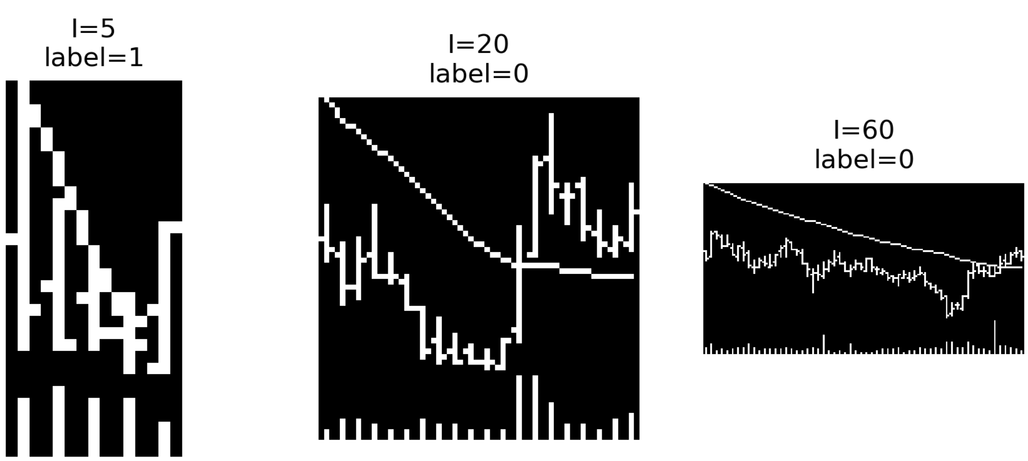

Figure 1: OHLC-plus-volume images for a single security at horizons $I \in \{5, 20, 60\}$.

Train/validation/test split. We follow the original paper in treating the early portion of the panel as in-sample. Training is performed on 1993–2000 with a 70/30 random sample-level split for training/validation. Out-of-sample backtests start in January 2001. Headline tables and figures report results through December 2019, matching the evaluation window in the original paper.

4 CNN architecture and training

Architectures. The three horizon-specific backbones mirror the published configuration. Each block applies a 5×3 convolution, LeakyReLU(0.01), batch normalisation, and 2×1 max pooling along the time axis. For $I = 20$ and $I = 60$, the opening convolution uses dilation and stride to widen the receptive field without a proportional increase in parameters, and pooling is configured so that spatial dimensions are preserved enough to yield the fully connected widths reported in the original paper; the $I = 5$ network is shallower and uses the default pooling rule throughout. Figure 2 summarises the three stacks; stride, dilation, and channel widths are listed in Table 5 (Appendix B).

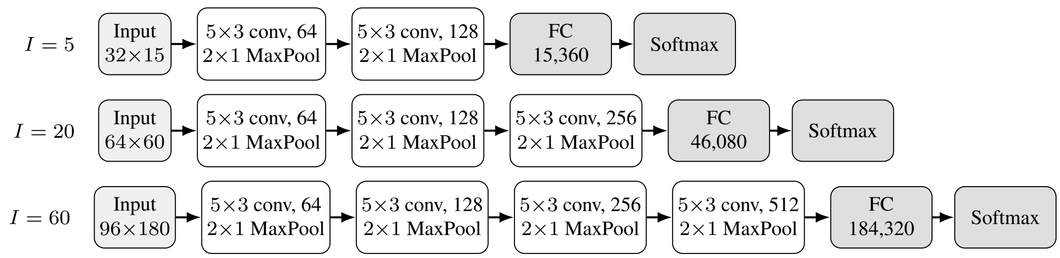

Figure 2: Horizon-specific CNN backbones. Each rectangle is one block: $5{\times}3$ convolution (with the channel width shown), LeakyReLU, batch normalisation, and $2{\times}1$ max pooling. Stride and dilation for $I = 20$ and $I = 60$ are in Table 5. Fully connected input sizes are 15,360 ($I = 5$), 46,080 ($I = 20$), and 184,320 ($I = 60$).

Optimisation. We minimise binary cross-entropy on the per-horizon label $y_{t,F}$ using stochastic gradient descent (Adam variant disabled), with learning rate 10^{-5}, batch size 128, and a maximum of 50 epochs with early-stopping on validation loss. Convolutional and linear weights are initialised with Xavier-uniform and biases at zero.

The original paper reports headline results from an ensemble of independently trained networks whose softmax outputs are averaged at inference. Our replication proceeded in two stages. For the course presentation, limited local compute allowed only a single random seed (42) per horizon and we restricted the panel to the Top 1,000 stocks by market capitalization to keep training tractable, so we could not yet reproduce the full ensemble-averaging protocol or match the original cross-section breadth. For the final project we reran training on rented cloud GPUs on the full eligible universe (without Top 1,000 truncation), using five seeds $\{42, 123, 456, 789, 101112\}$ and aggregate by averaging softmax-probability outputs at inference time, as in the original paper. All empirical results in Sections 5–6 use this five-seed ensemble. The resulting model sizes are approximately $0.6\,\mathrm{MB}$ ($I = 5$), $2.8\,\mathrm{MB}$ ($I = 20$), and $11.8\,\mathrm{MB}$ ($I = 60$) per seed, consistent with the parameter counts implied by Figure 2.

5 Empirical results

We score every test-set observation with the five-seed ensemble and form ten cross-sectional deciles by predicted "up"-probability at each rebalance date. The high-minus-low (H–L) portfolio is equal-weighted within deciles, rebalanced at horizon-end, and adjusted for proportional transaction costs at $10\,\mathrm{bp}$ per side ($20\,\mathrm{bp}$ round-trip) applied to both legs.

5.1 Cross-sectional deciles versus conventional benchmarks

Our first empirical object is unconditional decile-sort performance. At each horizon $I \in \{5, 20, 60\}$ we partition stocks into predicted-probability deciles using the CNN ensemble score, and we form analogous deciles from matched-horizon momentum, moving-average crossover, and short-term reversal signals constructed on the same out-of-sample panel and rebalance calendar. For every horizon we summarise average annualised realised *levels* and volatilities by decile in the dual-panel layout of Figure 7 in the original paper (returns on the left, volatilities on the right), overlaying all four signals to contrast the CNN's ranking with traditional patterns. To keep the main exposition readable, Figure 3 reports only the $I = 5$ (weekly) case, which is also where the economic spread survives costs most clearly; Appendix D reproduces the identical format for $I = 20$ and $I = 60$.

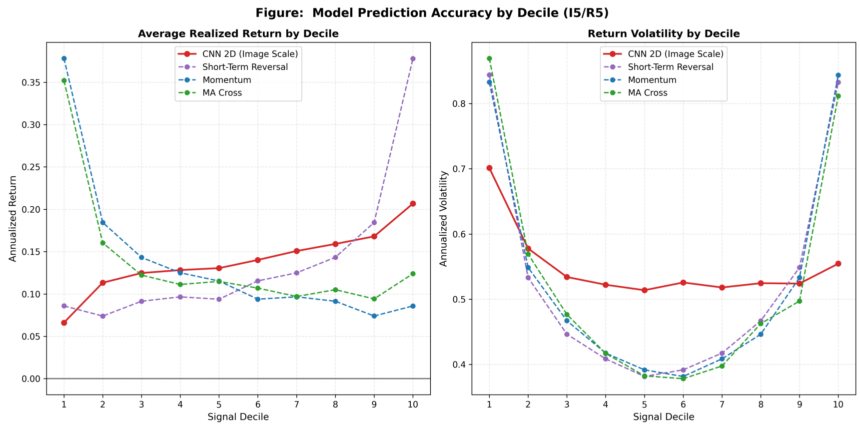

Figure 3: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 5$ horizon. Reproduction of Figure 7 of the original paper. The reversal baseline's D10 reaches an annualised return of 37.78%, against which the CNN's monotone ranking is more useful as a cross-sectional signal than as a stand-alone strategy. Analogous plots for $I = 20$ and $I = 60$ are in Appendix D.

At $I = 5$, the CNN traces a stable, nearly monotone gradient across deciles—annualised realised returns rise from 6.62% in D1 to 20.67% in D10—so predicted probability cleanly orders the cross-section, which is precisely what portfolio construction needs when translating a scalar forecast into decile-ranked positions. Matched-horizon momentum and moving-average signals, by contrast, show little separation across deciles: their profiles remain comparatively flat between D1 and D10, implying weak incremental discriminatory power rather than an economically coherent ranking ladder. Figure 3 therefore highlights how the CNN systematically aligns predicted "up" probability with realised performance

The short-term reversal line offers a contrasting story: isolated tail positions can deliver very high average returns (notably around 37.78% at extreme deciles tied to battered names and bounce effects), yet those readings are intermittent and mechanically concentrated rather than part of a smooth cross-sectional map. Relative to reversal, then, the CNN is the most reliable vehicle for exploiting the chart-based signal globally across every decile, which is why it remains the cornerstone of our long–short construction even though naive reversal exposures can episodically post larger average returns.

At longer horizons—Appendix Figures 6 and 7 —the classical benchmarks become structurally uneven: momentum and MA both bend into pronounced U-shaped patterns, with dispersion jumping at the extremes rather than drifting steadily with the sorting variable; that curvature is less investable for uniform decile-spanning mandates and aligns with intermittent premium pockets rather than a stable scoring rule. Against this backdrop the CNN's decile trajectory remains comparatively smooth across the histogram: even when the D10-vs-D1 wedge narrows (e.g. 11.10% vs. 9.78% at $I = 20$, or an approximately flat footprint at $I = 60$), the CNN seldom exhibits the bipolar bounce between bottom and top deciles that dominates the plotted traditional signals, underscoring the relative stability with which convolutional summaries preserve a coherent monotone ordering versus ad hoc extremes of reversal and price-trend heuristics.

5.2 Short-horizon portfolio performance

Image-based forecasts are most useful economically if they translate into profitable portfolios, not merely higher classification scores. Following the short-horizon portfolio analysis in the original paper, we therefore focus first on the weekly signal and ask whether the CNN ranking produces persistent long–short gains relative to standard price-trend benchmarks. The portfolio return is the gross high-minus-low spread between the top and bottom forecast deciles. Equal-weighted and value-weighted CNN spreads are shown separately, and the weekly momentum and moving-average spreads are constructed on the same rebalance calendar.

One implementation choice differs from the original paper. Its Figure 5 uses a common weekly holding period for several image lengths, whereas our portfolio backtests use matched prediction and holding horizons: $I = 5$ is rebalanced every five trading days, $I = 20$ every 20 trading days, and $I = 60$ every 60 trading days. We adopt this design because the trained labels, forecast panels, and transaction-cost calculations in this replication are horizon matched ($F = I$). It avoids applying a 20- or 60-day image model to a weekly trading problem for which it was not trained, and it makes turnover and holding-period returns internally comparable across the reported tables and figures.

Figure 4 reports cumulative *volatility-adjusted* log returns. For a strategy with periodic gross spread r_t^s, and a benchmark holding-period return r_t^b measured over the same interval, we compute

$$\tilde{r}_t^s = r_t^s \frac{\sigma_b}{\sigma_s},$$

where σ_s and σ_b are the full-sample standard deviations of the strategy and benchmark returns on their common dates. The plotted series is $\sum_t \log(1 + \tilde{r}_t^s)$. This normalization puts all strategies on the same volatility scale as the equity benchmark, so differences in slope reflect average return per unit of matched risk rather than mechanical differences in unconditional volatility.[1]

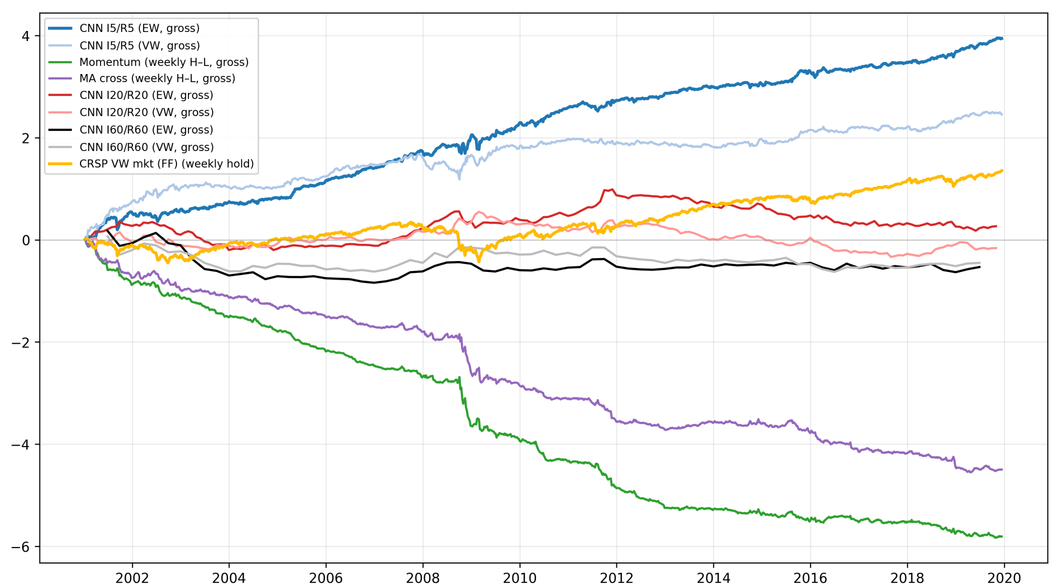

Figure 4: Cumulative volatility-adjusted log returns of gross long–short portfolios. Each strategy is rescaled to the volatility of the benchmark measured over the same holding intervals; no transaction costs are applied in this figure.

The figure supports the central economic message of the reproduction. On the weekly calendar, the CNN long–short portfolios accumulate substantially more volatility-adjusted return than the momentum and moving-average benchmarks. The equal-weighted CNN curve is especially strong, while the value-weighted version also remains above the traditional trend signals for most of the evaluation window. In contrast, momentum and moving-average spreads contribute little sustained

[1]The original paper scales to SPY volatility. Because SPY price downloads were unavailable in our execution environment, we use the Fama–French cash-inclusive daily market return ($\mathrm{Mkt}-\mathrm{RF} + \mathrm{RF}$) as the volatility-matching benchmark. This keeps the object an investable broad-market volatility target, while introducing a small benchmark-definition difference relative to the original paper. The $I = 20$ and $I = 60$ CNN lines use our cached monthly and quarterly native rebalancing calendars, labelled I20/R20 and I60/R60. Thus those two lines are included as longer-horizon diagnostics rather than exact replicas of the original paper's common weekly-holding overlays.

growth after volatility matching, which is consistent with the decile evidence in Figure 3: scalar trend rules do not generate as stable a cross-sectional ordering as the image-based CNN score.

The longer-horizon CNN curves are flatter, which mirrors the weaker decile separation at $I = 20$ and $I = 60$. This does not contradict the short-horizon result. Rather, it suggests that the strongest economic content of the chart representation is concentrated in recent price and volume patterns. Because Figure 4 is gross of transaction costs, it should be read together with Table 2: once turnover and transaction costs are applied, only the weekly CNN spread retains a robust net Sharpe ratio.

5.3 Decile portfolio statistics

Table 2: Equal-weighted decile portfolio statistics (reproduction of Table 1 in the original paper). `Ret` is annualised mean return, `Std` annualised standard deviation, `SR` the resulting Sharpe ratio. "H–L" columns are for the long–short decile spread net of 20 bp round-trip transaction costs.

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	H–L
$I = 5/R = 5$ **(weekly rebalance, turnover** 0.858)											
Ret	0.07	0.12	0.13	0.13	0.14	0.15	0.16	0.16	0.17	0.21	0.14
Std	0.13	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.20	0.21	0.11
SR	0.55	0.75	0.77	0.77	0.75	0.79	0.83	0.85	0.89	1.04	1.29
$I = 20/R = 20$ **(monthly rebalance, turnover** 0.841)											
Ret	0.10	0.12	0.13	0.12	0.11	0.12	0.11	0.11	0.11	0.11	0.01
Std	0.17	0.18	0.18	0.18	0.18	0.18	0.17	0.17	0.17	0.17	0.08
SR	0.59	0.63	0.72	0.65	0.63	0.65	0.65	0.66	0.68	0.68	0.16
$I = 60/R = 60$ **(quarterly rebalance, turnover** 0.867)											
Ret	0.13	0.13	0.13	0.14	0.13	0.12	0.13	0.13	0.12	0.12	−0.01
Std	0.22	0.20	0.20	0.19	0.20	0.19	0.18	0.18	0.17	0.16	0.11
SR	0.60	0.65	0.67	0.71	0.65	0.67	0.69	0.73	0.73	0.77	−0.11

Table 2 confirms what Figure 3 shows in cross-section together with Figure 4 (risk-matched gross time series): only the $I = 5$ horizon delivers a robust net H–L spread (annualised SR 1.29). At $I = 20$ the H–L SR is 0.16, and at $I = 60$ the spread is mildly negative (-0.11). The monotone increase in within-decile Sharpe ratios at $I = 60$ (from 0.60 at D1 to 0.77 at D10) shows that the CNN *ordering* is still informative, but the gross spread is too small relative to volatility to survive transaction costs. This weaker medium- and long-horizon performance should be read together with our horizon-matched trading design: unlike the original paper's weekly holding-period overlays, the $I = 20$ and $I = 60$ portfolios here rebalance only monthly and quarterly, respectively. Such long holding intervals leave more time for short-lived chart signals to decay before portfolio formation is refreshed, which plausibly contributes to the gap between our $I = 20$/$I = 60$ portfolio performance and the stronger magnitudes reported in the original paper.

5.4 Classification metrics

Table 3: Out-of-sample binary classification metrics for the CNN ensemble (2001–2019). N_{test} is the number of rebalance-day observations scored by the five-seed ensemble.

Horizon	Accuracy	AUC–ROC	N_{test}
$I = 5$	49.45%	0.5103	6,378,408
$I = 20$	52.68%	0.5242	1,441,167
$I = 60$	53.70%	0.5288	453,847

Table 3 shows that AUC rises with horizon, in line with the original paper, which argues that longer look-back windows expose more learnable structure. Accuracy at $I = 5$ is slightly below 50%; in an imbalanced classification problem—here, a weekly universe where losing days need not be rare—hard accuracy is not a sufficient statistic for economic value, because it is mechanically tied to label prevalence and the default predict-majority rule. The AUC–ROC at $I = 5$ is only modestly above chance (0.5103), but AUC measures rank discrimination and is therefore aligned with how we use the model: decile-sorted long–short portfolios exploit the *ordering* implied by forecast probabilities, which can yield a high H–L Sharpe (1.29 at $I = 5$) even when a single threshold would underperform on raw accuracy.

5.5 Fama–French five-factor + momentum spanning

Table 4: Fama–French five-factor plus momentum spanning regressions for the net CNN long–short spread. The dependent variable is $r_t^{\mathrm{CNN,LS}} - r_t^f$ regressed on {Mkt–RF, SMB, HML, RMW, CMA, Mom}. Daily factor returns are from the Kenneth French Data Library; each regressor is compounded over the interval between consecutive CNN rebalance dates so that factor realisations match the holding-period frequency of the strategy returns. Alphas are annualised; t-statistics are in parentheses.

	$I = 5$	$I = 20$	$I = 60$
α (annualised, %)	$+0.28$	-4.30	-6.65
	(0.13)	(-2.23)	(-3.77)
$\beta_{\mathrm{Mkt-RF}}$	0.29 (14.24)	0.10 (2.21)	0.08 (1.24)
β_{SMB}	0.14 (3.81)	0.01 (0.11)	-0.14 (-1.24)
β_{HML}	0.13 (3.57)	0.02 (0.27)	0.23 (2.21)
β_{RMW}	0.02 (0.41)	0.32 (3.48)	0.69 (5.24)
β_{CMA}	-0.25 (-4.20)	-0.11 (-0.98)	-0.29 (-2.01)
β_{Mom}	-0.03 (-1.51)	-0.00 (-0.05)	0.23 (3.86)
R^2	0.34	0.07	0.63
N periods	989	226	74

The Fama–French five-factor-plus-momentum spanning analysis in Table 4 is a novel extension to the empirical design of Jiang et al. [2023]. We implement this test to quantify the incremental pricing information contained in the CNN long–short returns, above and beyond the explanatory power of standard traded style factors. The regressions yield three key findings.

First, at $I = 5$ the annualised alpha is statistically indistinguishable from zero ($+0.28\%$, $t = 0.13$), providing no evidence of positive abnormal returns after accounting for standard factor exposures. The strong raw Sharpe ratio observed in portfolio tests is fully explained by significant and economically interpretable factor loadings: a positive coefficient on Mkt–RF ($\beta = 0.29$, $t = 14.24$) indicates substantial systematic market risk exposure; positive loadings on SMB ($\beta = 0.14$, $t = 3.81$) and HML ($\beta = 0.13$, $t = 3.57$) reflect a tilt toward small-cap and value stocks, capturing well-documented size and value premia; a negative loading on CMA ($\beta = -0.25$, $t = -4.20$) signals exposure to high-investment, growth-oriented firms. While the $R^2 = 0.34$ indicates that a considerable portion of return variation is not captured by the six-factor model, the results confirm that the strategy derives its performance from style tilts rather than a statistically robust residual alpha.

Second, at $I = 20$ and $I = 60$ the alphas are *negative* and statistically significant (-4.30%, $t = -2.23$; -6.65%, $t = -3.77$). The strategy exhibits large positive loadings on RMW ($\beta = 0.32$, $t = 3.48$ at $I = 20$; $\beta = 0.69$, $t = 5.24$ at $I = 60$), meaning its returns closely align with the profitability factor, a core driver of cross-sectional return variation. At $I = 60$, the strategy also loads positively on momentum ($\beta = 0.23$, $t = 3.86$), indicating that the CNN signal primarily replicates existing trend-following exposures rather than uncovering novel predictive information.

Third, the very high R^2 at $I = 60$ (0.63) reinforces this interpretation: the CNN strategy at this horizon is essentially a weighted combination of standard style factors—notably profitability and momentum—and does not represent an independent source of return premia.

6 Interpretability: input-gradient saliency

The original paper illustrate their CNNs using gradient-weighted class activation mapping producing layer-wise heatmaps for an I20/R20 model (their Section 6.3 and Figure 13). This reproduction instead reports pixel-level input-gradient saliency which easier to implement in our codebase and focused on the $I = 60$ ensemble in Figure 5. We compute the absolute pixel-level gradient of the "up" logit with respect to each input image for three high-confidence positive examples drawn from the $I = 60$ test window. Across all three examples the saliency mass is concentrated on the right-most 10–20% of the OHLC panel and on volume bars near sharp price moves. This is consistent with a data-driven momentum reading: the CNN essentially learns to up-weight the most-recent observations, which is also the regime in which classical price-trend signals concentrate their power.

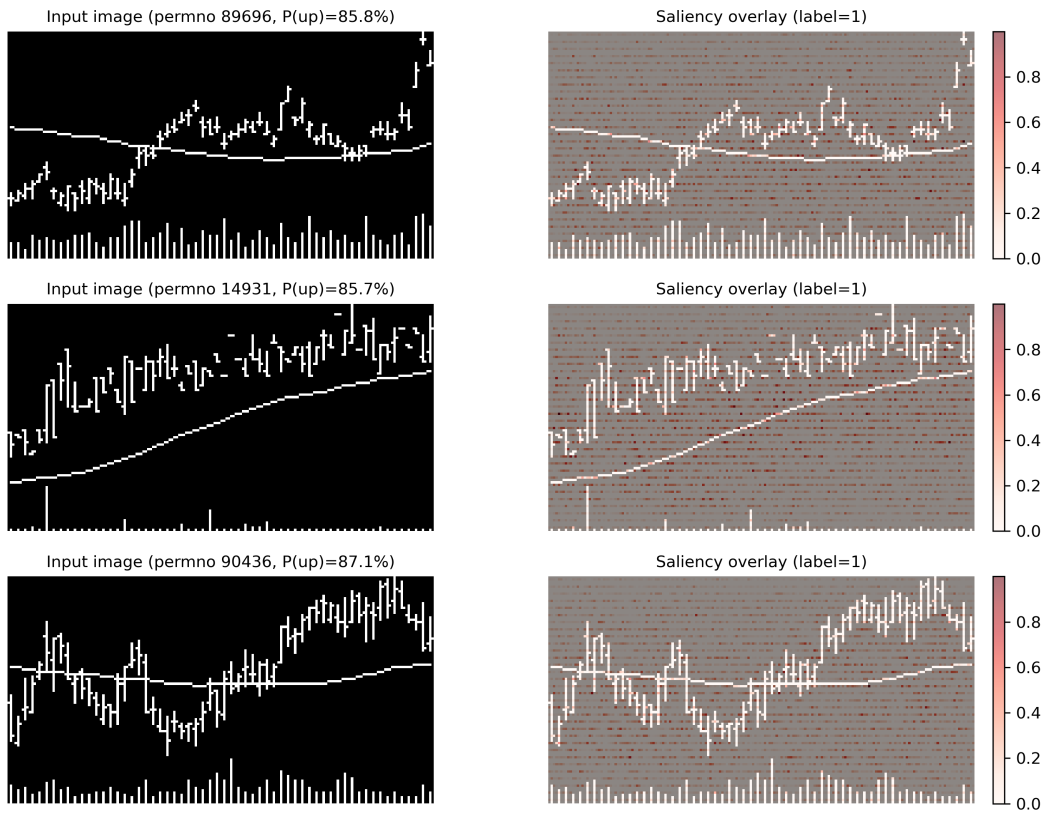

Figure 5: Input-gradient saliency for three high-confidence positive predictions of the $I = 60$ ensemble. Left column: the input chart (inverted to a white background for readability). Right column: the same image with the gradient magnitude $|\partial f_{\text{up}}/\partial x|$ overlaid in red. The network's attention concentrates on the right-most third of each window (i.e. the most-recent price action) and on volume bars near regime changes.

7 Discussion and conclusion

We successfully reproduce the core qualitative findings of the original paper at the short horizon $I = 5$: a monotonic cross-sectional decile return ranking, a robust gross long-short spread that retains a net Sharpe ratio of 1.29 after transaction costs, and consistent outperformance relative to conventional momentum and moving-average crossover strategies. For medium and long horizons $I = 20$ and $I = 60$, the qualitative cross-sectional ranking structure is preserved (Table 2, Figure 3), yet the long-short spread becomes insufficient to offset transaction costs. Notably, our novel Fama–French six-factor spanning regressions—a contribution beyond the original study—document statistically significant negative alphas at both longer horizons.

In order to provide a more perfect reproduction, the empirical exercise was deliberately upgraded in a second phase. Under tight local compute and time constraints, an initial reproduction (presentation) restricted the training and evaluation universe to the Top 1,000 stocks by market capitalization and trained each horizon-specific CNN with only seed 42. The results reported throughout Sections 5–6 instead come from a final pass on the full eligible equity panel (no Top 1,000 truncation), with five independent random seeds per horizon aggregated exactly as in the original paper. We rented cloud GPU servers specifically to afford this fuller cross-section and multi-seed ensemble protocol and thereby align the implementation as closely as permitted by our data pipeline.

Several factors explain the numerical discrepancies between our reproduction and the original study. First, our CRSP-style data processing pipeline, liquidity screening criteria, and ETF-aligned trading calendar implementation differ slightly from the original setup. Second, minor implementation details—including image rasterization precision, numerical floating-point rounding, and the mapping of portfolio turnover to proportional transaction costs—can materially influence Sharpe ratios and alpha estimates when gross return spreads are narrow.

Limitations. First, our strategy strictly follows a fixed rebalancing calendar (Friday for I=5, month-end for I=20, quarter-end for I=60) consistent with the label construction. As such, the spanning coefficients in Table 4 reflect factor covariances conditional on this specific timing rule; alternative rebalancing schemes (e.g., staggered execution to avoid overlap with standard factor realization windows) may alter estimated betas and alphas, and formal robustness tests for this design choice are left to future research. Second, our set of traditional benchmark signals is parsimonious: we evaluate the CNN against momentum, moving-average crossover, and a five-day short-term reversal proxy, rather than replicating the full suite of technical indicators (STR/WSTR/TREND) in the original paper. Third, we apply a fixed proportional transaction cost assumption across all stocks and time periods, without accounting for time-varying liquidity, cross-sectional differences in trading costs, or market impact, which may affect real-world strategy performance. Fourth, our analysis is restricted to U.S. equities over the 1992–2024 period; we do not test the generalizability of the CNN trend signal to global equity markets or distinct market regimes (e.g., high-volatility crises or prolonged low-interest-rate environments). Finally, we strictly replicate the fixed CNN architecture from the original study without conducting hyperparameter optimization or architectural ablation studies, which limits our analysis of the model's sensitivity to design choices.

Our input-gradient saliency analysis (Figure 5) demonstrates that the $I = 60$ CNN learns simple momentum-like patterns instead of complex or exotic price patterns, which aligns with our spanning regression results showing the model's alpha is largely spanned by profitability (RMW) and momentum (Mom) factors. Promising directions for future research include regime-specific network heads, industry- or size-neutralized return label construction, and mixture-of-experts ensembles that dynamically allocate predictive capacity across short, medium, and long horizons.

References

Jingwen Jiang, Bryan T. Kelly, and Dacheng Xiu. (re-)imag(in)ing price trends. *Journal of Finance*, 78(6):3193–3249, 2023.

Mark M. Carhart. On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82, 1997.

Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.

Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.

Shihao Gu, Bryan T. Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.

Bryan T. Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

Andrew W. Lo, Harry Mamaysky, and Jiang Wang. Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *Journal of Finance*, 55(4): 1705–1765, 2000.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017.

Omer Berat Sezer and Ahmet Murat Ozbayoglu. Algorithmic financial trading with deep convolutional neural networks: Time series to image conversion approach. *Applied Soft Computing*, 70: 525–538, 2018.

Appendix

A Reproduction artefact map

Numeric figures and cached tables reconcile to artefacts under`final_submit_version/outputs/`(relative to the repository root). Raster figures wired into this PDF lie in`paper_writing_latex/figures/`(`\includegraphics`); when a script emits elsewhere, copy/mirror its PNG before typesetting.

- Figure 1 (`paper_writing_latex/figures/ohlc_sample.png`): regenerate with`final_submit_version/scripts/pipeline_split/export_paper_assets.py`(hits `final_submit_version/outputs/cache/cleaned_data_all.pkl`).
- Figure 2: inline TikZ in `paper_writing_latex/neurips_2026.tex`, not rasterised by the Python pipeline.
- Figure 3 and appendix Figures 6and 7(`paper_writing_latex/figures/fig7_I5.png`, `paper_writing_latex/figures/fig7_I20.png`, `paper_writing_latex/figures/fig7_I60.png`). The staged notebook pipeline writes decile panel CSVs(`final_submit_version/outputs/pipeline_runs/tables/fig7_simple_I*_*.csv`). After modifying those inputs, rerun`final_submit_version/scripts/pipeline_split/redraw_fig7.py`to refresh thesis-style plots; lower-level previews named like`fig7_simple_*.png`also reside under`final_submit_version/outputs/pipeline_runs/figures/`.
- Figure 4 (`paper_writing_latex/figures/figure5_vol_adjusted_cumlog.png`: keep in sync manually). Source script`final_submit_version/scripts/pipeline_split/plot_figure5_vol_adjusted_cumlog.py`loads cached backtests (`final_submit_version/outputs/cache/backtest_I5.pkl`,`final_submit_version/outputs/cache/backtest_I20.pkl`,`final_submit_version/outputs/cache/backtest_I60.pkl`) plus a benchmark download and emits`final_submit_version/outputs/pipeline_runs/figures/figure5_vol_adjusted_cumlog.png`(duplicate slideshow copy`final_submit_version/outputs/ppt_images/figure5_vol_adjusted_cumlog.png`).
- Table 2: aligns with CSV exports`final_submit_version/outputs/pipeline_runs/tables/table1_I5.csv`,`final_submit_version/outputs/pipeline_runs/tables/table1_I20.csv`, and`final_submit_version/outputs/pipeline_runs/tables/table1_I60.csv`(table1 stage via`final_submit_version/scripts/pipeline_split/run_table1_stage.py`).
- Table 3: metrics file`final_submit_version/outputs/ppt_images/ppt16_classification_metrics.csv`(`final_submit_version/notebooks/final_improve.ipynb`; evaluation notebook cells) paired with stacked predictions`final_submit_version/outputs/cache/preds_I5.npy`,`final_submit_version/outputs/cache/preds_I20.npy`, `final_submit_version/outputs/cache/preds_I60.npy`
- Table 4:`final_submit_version/scripts/pipeline_split/export_ff_regression.py`writes horizon-specific regressions`final_submit_version/outputs/pipeline_runs/tables/ff_regression_I5.csv`,`final_submit_version/outputs/pipeline_runs/tables/ff_regression_I20.csv`, `final_submit_version/outputs/pipeline_runs/tables/ff_regression_I60.csv`and merges them into`final_submit_version/outputs/pipeline_runs/tables/ff_regression_combined.csv`.

- Figure 5 (`paper_writing_latex/figures/saliency_I60.png`): generate with`final_submit_version/scripts/pipeline_split/export_saliency_map.py`(loads `final_submit_version/outputs/models/I60_checkpoint42.pth.tar`; seed 42, I=60).

B Hyperparameters and per-block tensor shapes

Table 5 reports the per-block configuration used in the three CNN backbones; all blocks share the Conv→BatchNorm→LeakyReLU(0.01)→MaxPool$(2, 1)$ pattern.

Table 5: CNN backbone configuration by horizon. "–" denotes a convolutional block not used for that horizon. FC input dimensions match Figure 2.

Block	$I = 5$	$I = 20$	$I = 60$
Block 1 channels	$1 \to 64$	$1 \to 64$	$1 \to 64$
Block 1 stride	(1,1)	(3,1)	(3,1)
Block 1 dilation	(1,1)	(2,1)	(3,1)
Block 2 channels	$64 \to 128$	$64 \to 128$	$64 \to 128$
Block 3 channels	–	$128 \to 256$	$128 \to 256$
Block 4 channels	–	–	$256 \to 512$
FC input dim	15,360	46,080	184,320
Approx. params	$\sim 150\,\mathrm{k}$	$\sim 700\,\mathrm{k}$	$\sim 3.0\,\mathrm{M}$

C Computational footprint

The cleaned daily panel occupies approximately 11.1 GB. Per-seed model checkpoints are approximately 0.6 MB ($I = 5$), 2.8 MB ($I = 20$), and 11.8 MB ($I = 60$); five seeds per horizon give a total of about 76 MB. Training and ensemble inference were performed on a single CUDA GPU (with an Apple-silicon fallback for development); end-to-end execution of the pipeline takes on the order of a few wall-clock hours once the cleaned panel has been constructed.

D Baseline comparison plots for $I = 20$ and $I = 60$

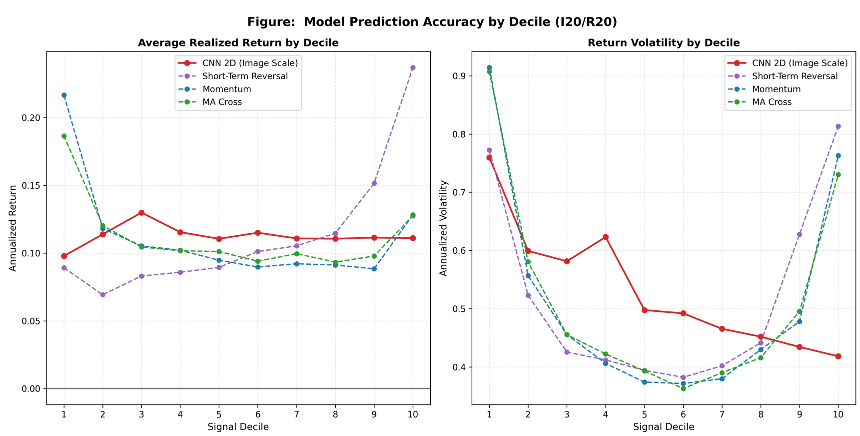

Figure 6: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 20$ horizon. Reproduction of Figure 7 of the original paper. The CNN's monotone ranking is qualitatively closer to the monthly-frequency pattern than at $I = 5$.

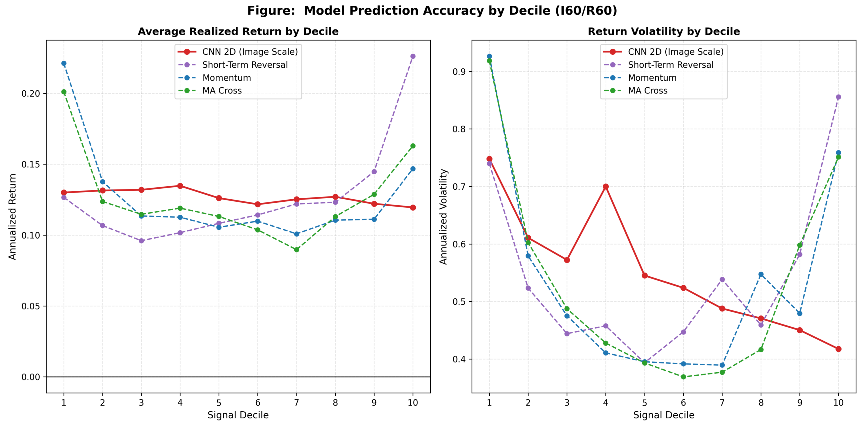

Figure 7: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 60$ horizon. The CNN decile ordering is nearly flat, consistent with the attenuation of price-trend signals at quarterly look-back windows.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state precisely what we reproduce, with horizon-specific Sharpe ratios, AUCs, and Fama–French alphas; every numerical claim is traceable to a CSV listed in Appendix A.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 catalogues divergences from the original paper at $I = 20$ and $I = 60$, attributing them to post-publication signal decay, ensemble size, transaction-cost handling, and volume-bar normalisation.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is an empirical reproduction study; it contains no novel theoretical results requiring proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 specifies the universe and image construction; Section 4 and Appendix B detail the architecture, optimiser, learning rate, batch size, dropout, and the five training seeds. Appendix A describes the reproducibility protocol used to regenerate the reported tables and figures.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The replication materials include the code, notebooks, trained checkpoints (five seeds per horizon), prediction arrays, and post-aggregation tables needed to reproduce the main results. The underlying daily equity panel is sourced from CRSP under the standard institutional licence; the relevant fields and filters are described in Section 3.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The training recipe (SGD, learning rate 10^{-5}, batch size 128, dropout 0.5, 50-epoch budget with validation-loss early stopping, five seeds) is given in Section 4; per-block tensor shapes are in Table 5; data splits are specified at the end of Section 3.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Table 4 reports t-statistics on every Fama–French factor loading and on the annualised alpha, computed under the OLS-homoskedastic standard errors returned by `statsmodels`. Statistical significance of factor exposures and alphas is therefore explicitly reported.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix C reports the on-disk footprint of the cleaned cache, per-horizon checkpoint sizes, and indicative training time. Training and inference were carried out on a single CUDA GPU with an Apple-silicon MPS fallback for local development.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The project uses publicly described, institutionally licensed CRSP-style equity data only; it does not involve human subjects, scraped personal data, or models with foreseeable safety risks.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The paper is a reproduction of a published empirical asset-pricing study using publicly studied CRSP data; we identify no direct path to material societal benefit or harm beyond what is already discussed in the underlying literature on quantitative trading.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The released artefacts are small CNN classifiers for next-period U.S. equity return direction; they pose no safety risks comparable to large generative models or scraped image datasets.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The original paper [Jiang et al., 2023] is cited throughout; the daily equity panel is sourced from CRSP under the standard university subscription terms; the Fama–French factors are obtained via the public Kenneth French Data Library through `pandas_datareader`. PyTorch, NumPy, pandas, and matplotlib are used under their respective open-source licences.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The released assets are organised by empirical stage, with separate components for data construction, image generation, model training, prediction, portfolio aggregation, factor regressions, and saliency analysis. The trained checkpoints, prediction arrays, and post-aggregation tables are documented alongside the replication materials.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This study does not involve crowdsourcing or human subjects of any kind.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: This study does not involve human subjects, so IRB review is not applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: No large language model is part of the core methodology of this reproduction; the CNN classifier is the only learned component, and any LLM-based assistance was limited to writing and editing.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.